



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune
Subject: Computer Vision Laboratory

Experiment - 4

Aim: To manually compute and visualise Harris corner responses in grayscale images.

Software Tools Required: VS Code, Python 3.12

Theory:

The Harris Corner Detector is a popular technique for finding corners—distinct points where intensity changes sharply in both directions—in grayscale images. Here's how it works and the role of Sobel operators and kernel sizes:

1. Grayscale Conversion

First, images are converted to grayscale so that the process operates on single-channel intensity data.

2. Gradient Computation (Sobel Operator)

To detect intensity changes, gradients in the x (I_x) and y (I_y) directions are calculated. The Sobel operator is commonly used for this step—it applies a convolutional kernel (filter) to approximate the derivatives:

- The kernel size (typically 3x3, but can be larger) affects the smoothness and accuracy of the gradient:
 - Smaller kernels (3x3): More sensitive to fine details, but also noise.
 - Larger kernels (e.g., 5x5, 7x7): Smoother, less sensitive to noise but may blur fine features.

3. Structure Tensor Formation

For each pixel, construct a 2x2 matrix (the "structure tensor" or "second moment matrix") using the squared gradients and their product. These values are summed or averaged over a window (kernel) around each pixel, with the window size also adjustable (e.g., 3x3, 5x5, 7x7):

$$M = \begin{bmatrix} \langle I_x^2 \rangle & \langle I_x I_y \rangle \\ \langle I_x I_y \rangle & \langle I_y^2 \rangle \end{bmatrix}$$

The size of the window kernel determines the level of local neighborhood considered for corner detection.



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4. Harris Response Calculation

The Harris corner response is computed as:

$$R = \det(M) - k[\text{trace}(M)]^2$$

where k is a sensitivity parameter (typically 0.04–0.06). Large positive values of R indicate strong corners.

5. Thresholding and Non-Maximum Suppression

Threshold the response values to identify significant corners, then apply non-maximum suppression to localize them precisely.

6. Visualization

Detected corners are usually marked visually (e.g., with circles or crosses) over the original image.

Summary of Sobel and Kernel Roles:

- The Sobel operator provides the image gradients required for Harris detection.
- The kernel size for Sobel determines gradient detail versus noise suppression.
- The averaging kernel/window in the structure tensor computation controls how much neighborhood information influences corner strength.

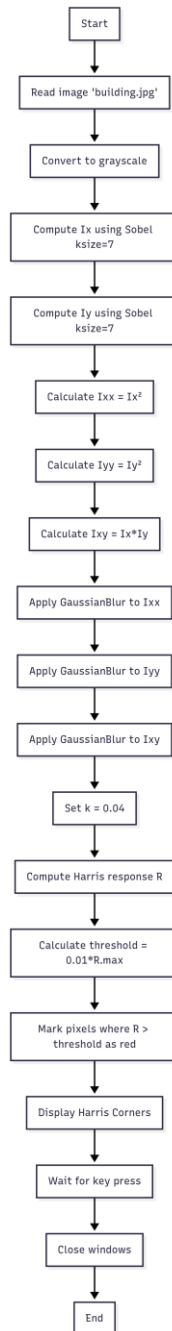
Fine-tuning these kernel sizes is important: smaller values increase sensitivity (but risk noise), while larger sizes smooth responses (possibly missing detailed corners). Adjusting both allows for optimal detection depending on the application and image quality.



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Flowchart



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Program

```
C: > Users > romil > Downloads > import numpy as np.py > ...
1  import numpy as np
2  import cv2
3
4  img = cv2.imread(r"C:\Users\romil\Downloads\cv\building.jpg")
5  if img is None:
6      print("Error: Image not found or cannot be opened!")
7      exit()
8  gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
9  Ix = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
10 Iy = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
11 Ixx = Ix ** 2
12 Iyy = Iy ** 2
13 Ixy = Ix * Iy
14 Ixx = cv2.GaussianBlur(Ixx, (5, 5), 0)
15 Iyy = cv2.GaussianBlur(Iyy, (5, 5), 0)
16 Ixy = cv2.GaussianBlur(Ixy, (5, 5), 0)
17 k = 0.04
18 R = (Ixx * Iyy - Ixy ** 2) - k * (Ixx + Iyy) ** 2
19 threshold = 0.01 * R.max()
20 img[R > threshold] = [0, 0, 255]
21 cv2.imshow('Harris Corners', img)
22 cv2.waitKey(0)
23 cv2.destroyAllWindows()
24
```

Input Image



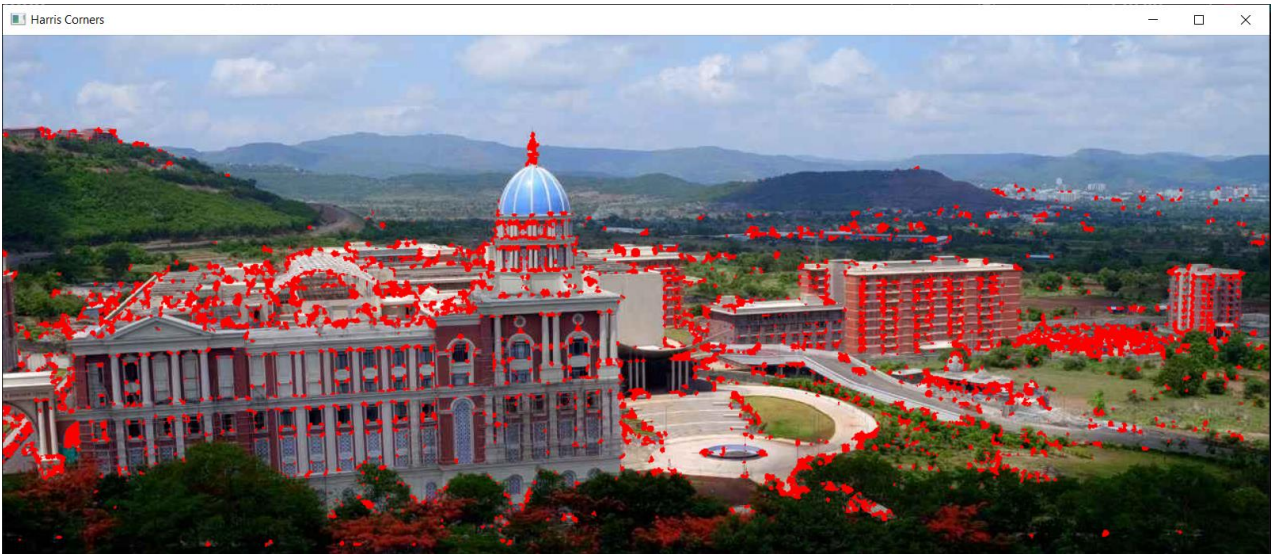
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Output Image



Conclusion

The Harris Corner Detection algorithm effectively identifies corner points in images by analyzing intensity gradients and their spatial derivatives. The implementation uses Sobel operators to compute gradients, applies Gaussian smoothing to reduce noise, and calculates a corner response function that highlights regions where intensity changes occur in multiple directions. By applying appropriate thresholding to the Harris response, the algorithm successfully marks corner points in red, providing a foundation for feature detection in computer vision applications like object recognition, image matching, and 3D reconstruction.